(b) Values of n, $\frac{1}{n}$ and I are given in Table 2.1.

Table 2.1

n	$\frac{1}{n}$	$I\,/\,\mu A$	$\frac{1}{I}\,/\,10^3 A^{-1}$
5	0.200	455 ± 5	
6	0.167	525 ± 5	
7	0.143	580 ± 5	
8	0.125	635 ± 5	
9	0.111	685 ± 5	
11	0.0909	765 ± 5	

Calculate and record values of $\frac{1}{I}\,/\,10^3 A^{-1}$ in Table 2.1. Include the absolute uncertainties in $\frac{1}{I}$. [2]

(c) **(i)** Plot a graph of $\frac{1}{I}\,/\,10^3 A^{-1}$ against $\frac{1}{n}$. Include error bars for $\frac{1}{I}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ... [2]